Introduction to Deep Learning

Caltech EE/CS/CNS 148a - Winter 2026
P. Perona

Lecture 0

Notation, diagrams etc.

Notation

$(x_n, y_n) \in \mathscr{D}$	data samples in dataset $\mathscr{D}$
$n = 1,2,\ldots,N$	data index
$\ell = 0,1,\ldots,L$	layer index
$x^{(\ell)}$	neurons' activation for layer ℓ
$x^{(0)} = x$	input to the network
$x^{(L)} = \hat{y}$	output of the network
$i, j = 0,\ldots M_i, M_j$	neuron indices (0 for bias neuron)
$W_{ij}^{(\ell)}$	weight from input j to output i a layer ℓ
$\sigma(\cdot)$	nonlinearity
$s^{(\ell)} = W^{(\ell)} x^{(\ell-1)}$	sum of inputs
$t = 0,1,\ldots,T$	optimization step
$k = 1,2,\ldots,K$	class index
$b = 1,2,\ldots,B$	mini-batch index
$c = 1,2,\ldots,C$	channel index
u, v	spatial indices for convolutions
$h = 1,2,\ldots,H$	attention head
$p = 1,2,\ldots,P$	parameter index

Multi-layer perceptron

Standard

$$x^{(0)} = x$$
$$s^{(\ell)} = W^{(\ell)} x^{(\ell-1)}, \qquad \ell = 1,\ldots,L$$
$$x^{(\ell)} = \sigma\left(s^{(\ell)}\right), \qquad \ell = 1,\ldots,L-1$$
$$x^{(L)} = \hat{y}$$

$\longleftrightarrow$

Standard w indices

$$x_j^{(0)} = x_j$$
$$s_i^{(\ell)} = \sum_j W_{ij}^{(\ell)} x_j^{(\ell-1)}$$
$$x_i^{(\ell)} = \sigma\left(s_i^{(\ell)}\right), \qquad \ell = 1,\ldots,L-1$$
$$x^{(L)} = \hat{y}$$

Soft-max

$$x^{(0)} = x$$
$$s^{(\ell)} = W^{(\ell)} x^{(\ell-1)}, \qquad \ell = 1,\ldots,L$$
$$x^{(\ell)} = \sigma\left(s^{(\ell)}\right), \qquad \ell = 1,\ldots,L-1$$
$$\hat{y}_k = \frac{e^{s_k^{(L)}}}{\sum_{j=1}^{K} e^{s_j^{(L)}}}$$

Backpropagation

$$\mathscr{L} = \ell(\hat{y}, y)$$
$$\delta^{(L)} = \nabla_{s^{(L)}} \mathscr{L}$$
$$\delta^{(\ell)} = \left(W^{(\ell+1)}\right)^\top \delta^{(\ell+1)} \odot \sigma'\left(s^{(\ell)}\right), \qquad \ell = L-1,\ldots,1$$
$$\frac{\partial \mathscr{L}}{\partial W^{(\ell)}} = \delta^{(\ell)} \left(x^{(\ell-1)}\right)^\top, \qquad \ell = 1,\ldots,L$$

Matrix-vector notation

$$\mathscr{L} = \ell(\hat{y}, y)$$
$$\delta_i^{(L)} = \frac{\partial \mathscr{L}}{\partial s_i^{(L)}}$$
$$\delta_i^{(\ell)} = \sum_j W_{ji}^{(\ell+1)} \delta_j^{(\ell+1)} \sigma'\left(s_i^{(\ell)}\right), \qquad \ell = L-1,\ldots,1$$
$$\frac{\partial \mathscr{L}}{\partial W_{ij}^{(\ell)}} = \delta_i^{(\ell)} x_j^{(\ell-1)}$$

All indices shown